Molecular Structure

The Born–Oppenheimer approximation: nuclei are treated as stationary while electrons move in their field.

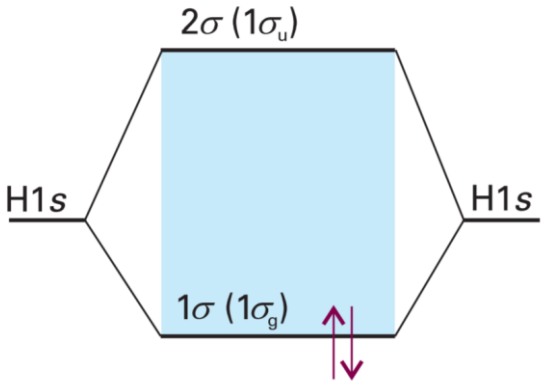
Valence-bond theory (localized)

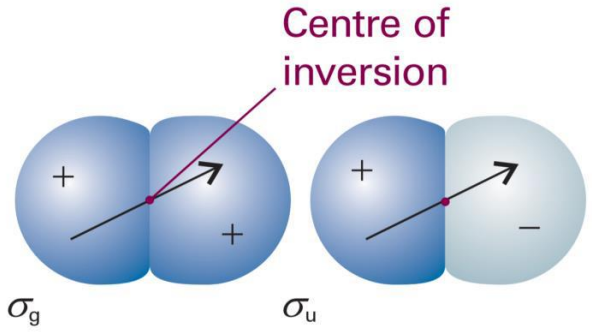
Molecular orbital theory (delocalized) $\psi_{\pm} = N(A \pm B)$

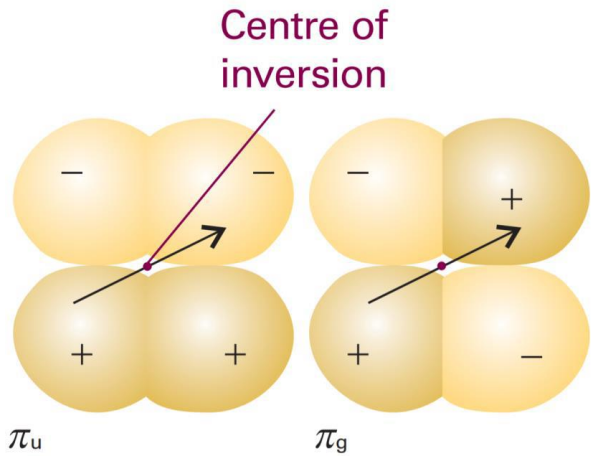
LCAO-MO: An approximate molecular orbital formed from a linear combination of atomic orbitals

Bond Order $b = \frac{1}{2}(n - n^{*})$

MO energy level diagrams



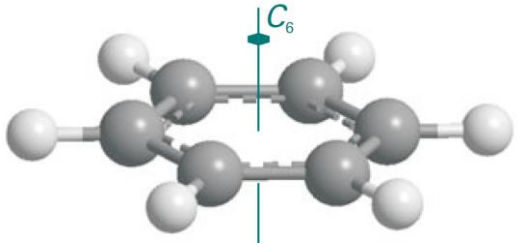




Electronic Configuration H_2: $1\sigma_g^2$

For homonucleardiatomic molecules

Symmetry elements and symmetry operations



1.***n*-fold rotation C_n** : A rotation through 360°/*n*

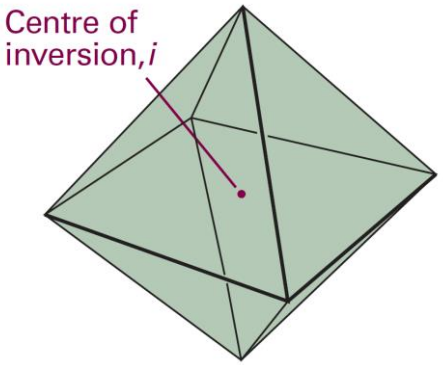
2. A **reflection** in a **mirror plane**,σ

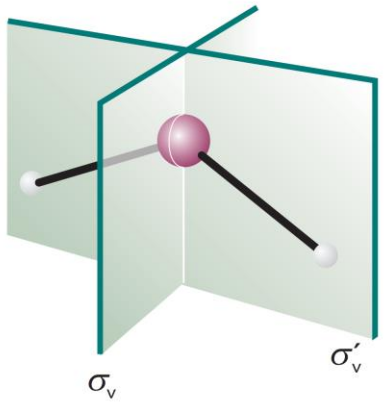
Vertical mirror plane, σ_V

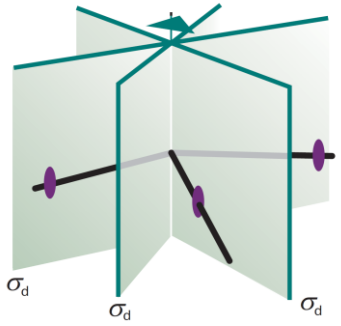
Dihedral mirror planes, σ_d

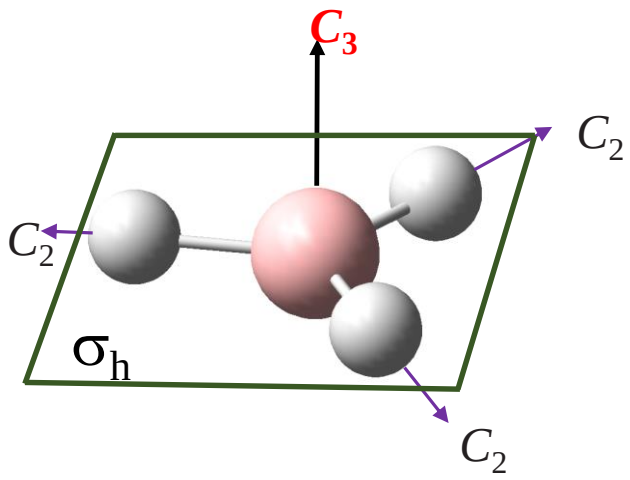
Horizontal mirror planes, σ_h

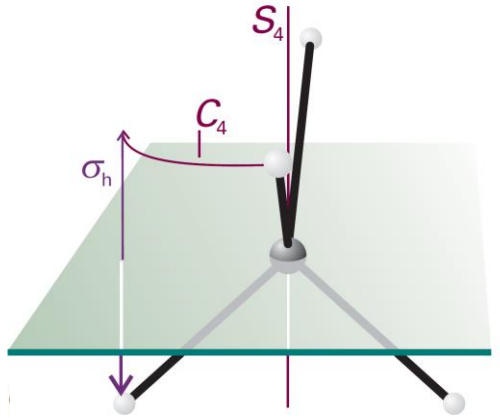
3. **Inversion** through a **center of symmetry**, ***i***











4. ***n*-fold improper rotation** about an ***n*-fold improper rotation axis, S_n** $S_n = \sigma_h C_n$

The point groups

C_1, no element other than the identity E

C_i, has the identity E and the inversion alone

C_s, has the identity E and a mirror plane alone

C_n, E + a C_n axis

C_{nv}, E + a C_n axis + n σ_V

C_{nh}, E + a C_n axis + a σ_h

T: 4 C_3, 3 C_2 (**T_h**: T + 3σ_h) (**T_d**: T + 3S_4)

O: 4 C_3, 3 C_4 (**O_h**: O + 3σ_h)

I: 6 C_5, 10 C_3 (**I_h**: I + i)

D_n, E + a C_n principal axis + n C_2 axes $\perp$ C_n

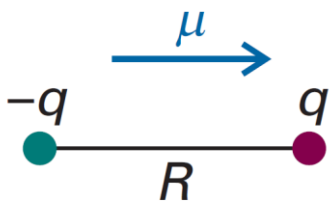
D_{nh}, E + a C_n principal axis + n C_2 axes $\perp$ C_n + a σ_h

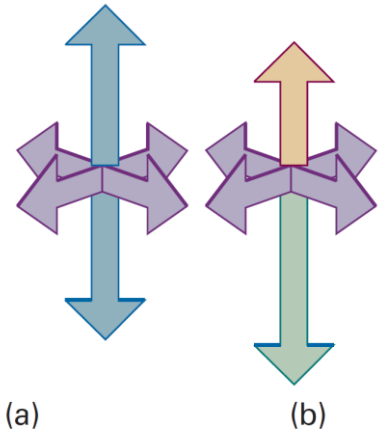
D_{nd}, E + a C_n principal axis + n C_2 axes $\perp$ C_n + n σ_d

S_n has **a** S_n improper rotation axis, n=2m, m >1

Some immediate consequences of symmetry

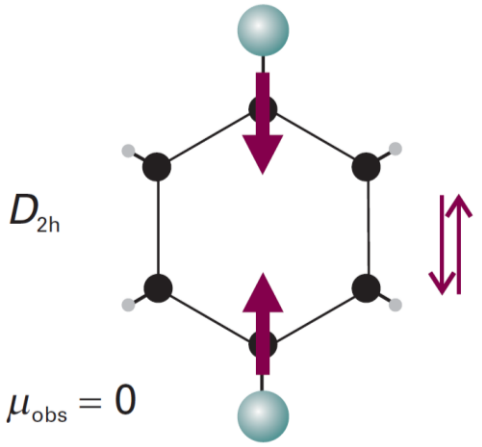
Polarity

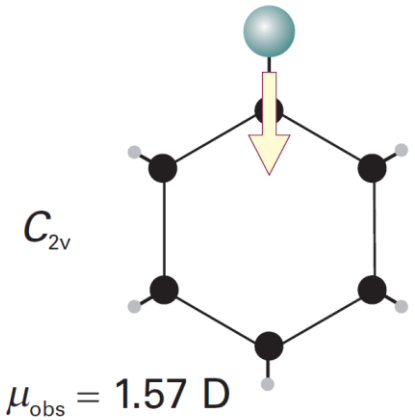


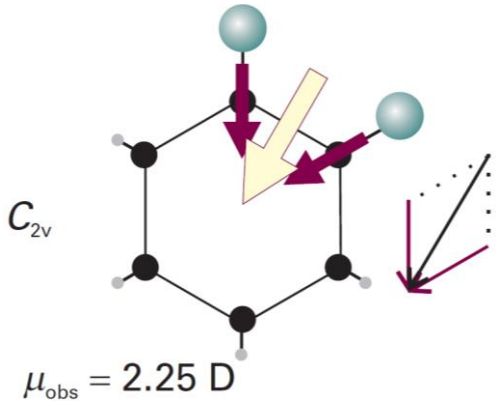


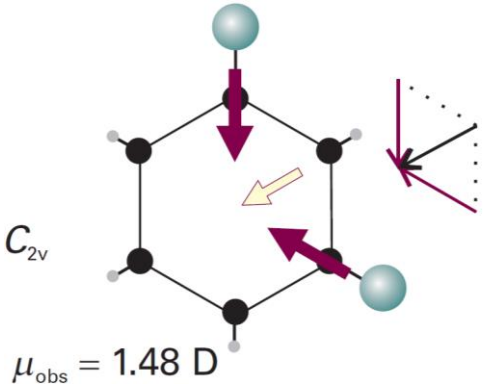
Only molecules with C_n , C_{nv}, and C_s symmetries may have a permanent electric dipole moment.

For C_n and C_{nv}, that dipole moment must lie along the symmetry axis.









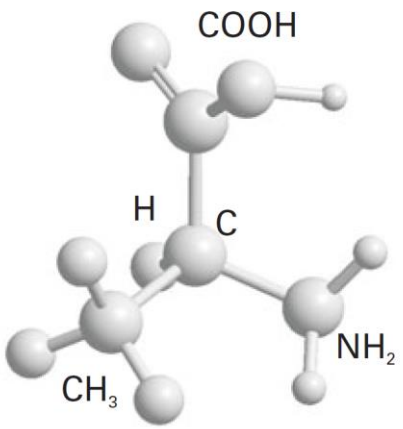
Some immediate consequences of symmetry

Chirality

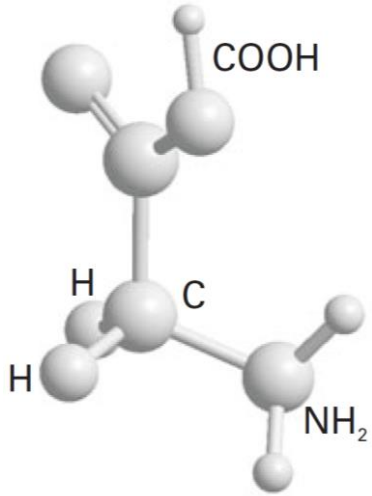
A **chiral (achiral) molecule:** cannot (can) be superimposed on its mirror image.

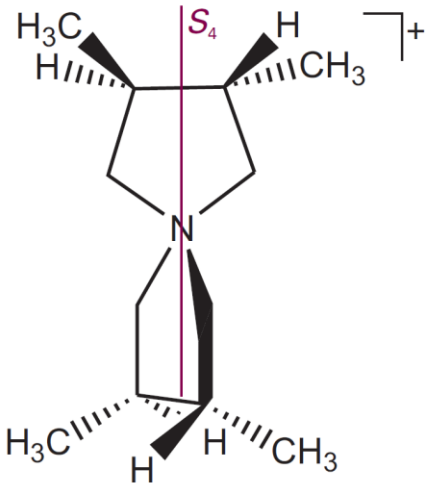
A molecule may be chiral, only if it does not possess an S_n axis. ⟷

A molecule may be chiral, if it does not have a center of inversion or a mirror pane, or an S_4 axis.



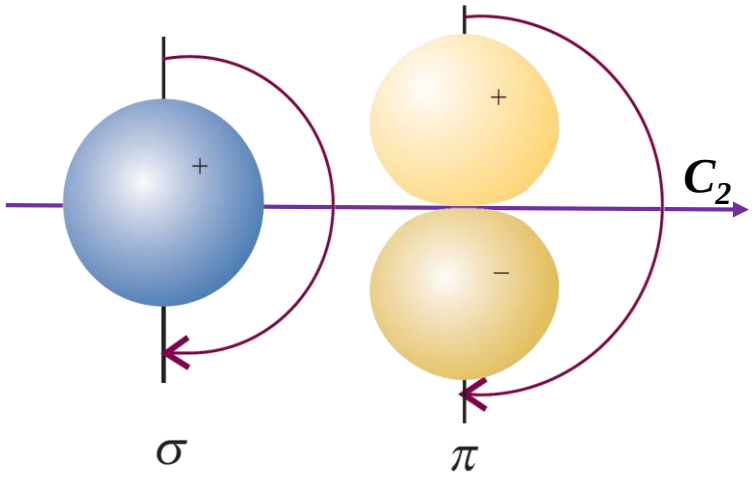
Chiral





Achiral

Representation of the group (群的表示)



$C_2\sigma = +1\sigma$

$C_2\pi = -1\pi$

The labels to denote the symmetries of orbitals (a, a_1, e, e_g…) indicate the behavior of the orbitals under the symmetry operations of the relevant point group of the molecule.

The **character table (特征标表)** of the group: a table that characterizes the different symmetry types possible in the point group.

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	−1
E	2	−1	0

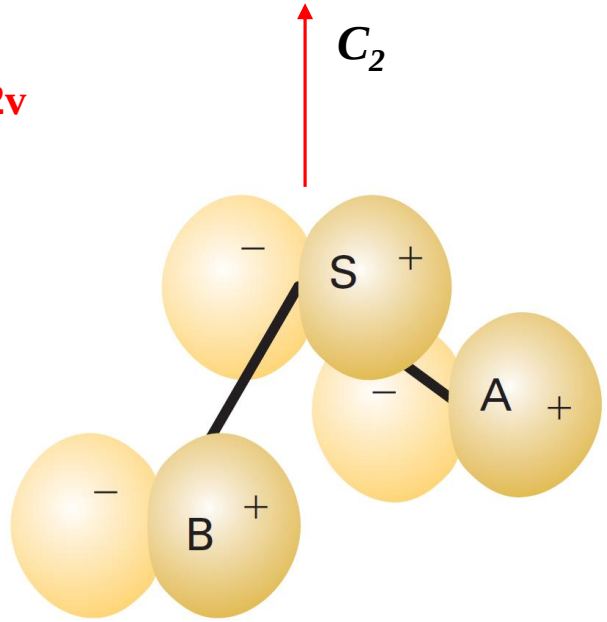
Symmetry species; Labels for irreducible representation

Symmetry operations

C_{2v}	E	C_2	σ_v	σ'_v
A_1	1	1	1	1
A_2	1	1	−1	−1
B_1	1	−1	1	−1
B_2	1	−1	−1	1

Characters, χ

C_{2v}



$$(p_S, p_B, p_A) \xleftarrow{\sigma_v} (p_S, p_A, p_B)$$

$$(p_S, p_B, p_A) = (p_S, p_A, p_B)\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = (p_S, p_A, p_B)\boldsymbol{D}(\sigma_v)$$

A basis

A representative of the operation

$$(-p_S, -p_B, -p_A) \xleftarrow{C_2} (p_S, p_A, p_B)$$

$$(-p_S, -p_A, -p_B) \xleftarrow{\sigma'_v} (p_S, p_A, p_B)$$

$$\boldsymbol{D}(E) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\boldsymbol{D}(C_2) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\boldsymbol{D}(\sigma'_v) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Representatives take different forms according to the basis.

Basis (p_S, p_A, p_B)

$$\boldsymbol{D}(E)=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix} \quad \boldsymbol{D}(C_2)=\begin{pmatrix}-1&0&0\\0&0&-1\\0&-1&0\end{pmatrix} \quad \boldsymbol{D}(\sigma_v)=\begin{pmatrix}1&0&0\\0&0&1\\0&1&0\end{pmatrix} \quad \boldsymbol{D}(\sigma_v')=\begin{pmatrix}-1&0&0\\0&-1&0\\0&0&-1\end{pmatrix}$$

Matrix representation, Γ

A 3-D representation: $\Gamma^{(3)}$

Character χ: the sum of diagonal elements of the representative of the operation

$\boldsymbol{D}(E)$	$\boldsymbol{D}(C_2)$	$\boldsymbol{D}(\sigma_v)$	$\boldsymbol{D}(\sigma_v')$
3	−1	1	−3

Basis (p_S, p_A, p_B)

$$\boldsymbol{D}(E) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad \boldsymbol{D}(C_2) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix} \qquad \boldsymbol{D}(\sigma_v) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \qquad \boldsymbol{D}(\sigma'_v) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

The symmetry operations of C_{2v} never mix p_S with p_A and p_B!

p_S itself is a basis for a 1-D representation $\Gamma^{(1)}$

$$\boldsymbol{D}(E) = 1 \qquad \boldsymbol{D}(C_2) = -1 \qquad \boldsymbol{D}(\sigma_v) = 1 \qquad \boldsymbol{D}(\sigma'_v) = -1$$

p_A and p_B are a basis for the 2-D representation $\Gamma^{(2)}$

$$\boldsymbol{D}(E) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad \boldsymbol{D}(C_2) = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \qquad \boldsymbol{D}(\sigma_v) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \boldsymbol{D}(\sigma'_v) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

Basis (p_S, p_A, p_B)

$$\boldsymbol{D}(E)=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix} \quad \boldsymbol{D}(C_2)=\begin{pmatrix}-1&0&0\\0&0&-1\\0&-1&0\end{pmatrix} \quad \boldsymbol{D}(\sigma_v)=\begin{pmatrix}1&0&0\\0&0&1\\0&1&0\end{pmatrix} \quad \boldsymbol{D}(\sigma_v')=\begin{pmatrix}-1&0&0\\0&-1&0\\0&0&-1\end{pmatrix}$$

Reducible representation 可约表示 $\boldsymbol{\Gamma^{(3)} = \Gamma^{(1)} + \Gamma^{(2)}}$

p_S itself is a basis for a 1-D representation $\Gamma^{(1)}$ ⟶ **Irreducible representation 不可约表示**

$$\boldsymbol{D}(E)=1 \quad \boldsymbol{D}(C_2)=-1 \quad \boldsymbol{D}(\sigma_v)=1 \quad \boldsymbol{D}(\sigma_v')=-1$$

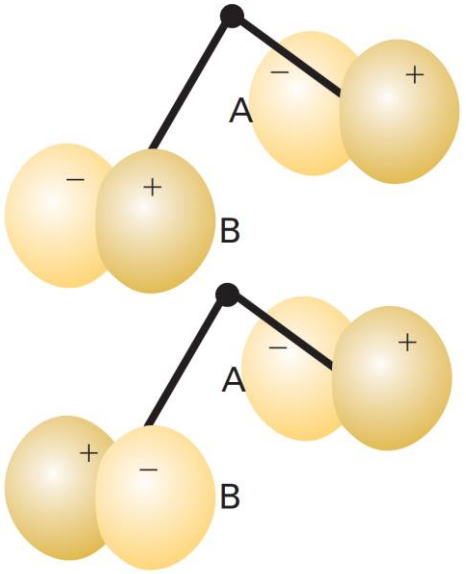
p_A and p_B are a basis for the 2-D representation $\Gamma^{(2)}$ ⟶ **Reducible representation 可约表示**

$$\boldsymbol{D}(E)=\begin{pmatrix}1&0\\0&1\end{pmatrix} \quad \boldsymbol{D}(C_2)=\begin{pmatrix}0&-1\\-1&0\end{pmatrix} \quad \boldsymbol{D}(\sigma_v)=\begin{pmatrix}0&1\\1&0\end{pmatrix} \quad \boldsymbol{D}(\sigma_v')=\begin{pmatrix}-1&0\\0&-1\end{pmatrix}$$

p_A and p_B are a basis for the 2-D representation $\Gamma^{(2)}$ ⟶ **Reducible representation 可约表示**

$$D(E)=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \quad D(C_2)=\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix} \quad D(\sigma_v)=\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} \quad D(\sigma_v')=\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}$$

A new basis, ($p_A + p_B$, $p_A - p_B$)



$$D(E)=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \quad D(C_2)=\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix} \quad D(\sigma_v)=\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} \quad D(\sigma_v')=\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}$$

$\Gamma^{(2)} = \Gamma^{(1)} + \Gamma^{(1)'}$

$p_A + p_B$, same as p_S, is a basis for the $\Gamma^{(1)}$

$$D(E)=1 \quad D(C_2)=-1 \quad D(\sigma_v)=1 \quad D(\sigma_v')=-1$$

$p_A - p_B$ is a basis for the $\Gamma^{(1)'}$

$$D(E)=1 \quad D(C_2)=1 \quad D(\sigma_v)=-1 \quad D(\sigma_v')=-1$$

$p_A + p_B$, same as p_S, is a basis for the $\Gamma^{(1)}$

$D(E) = 1$ $D(C_2) = -1$ $D(\sigma_v) = 1$ $D(\sigma_v') = -1$

$p_A - p_B$ is a basis for the $\Gamma^{(1)'}$

$D(E) = 1$ $D(C_2) = 1$ $D(\sigma_v) = -1$ $D(\sigma_v') = -1$

C_{2v}	E	C_2	σ_v	σ_v'
A_1	1	1	1	1
A_2	1	1	−1	−1
B_1	1	−1	1	−1
B_2	1	−1	−1	1

A or B: denote a 1-D representation.
Under the principal rotation,
$\chi = +1 \rightarrow$ A; $\chi = -1 \rightarrow$ B.
E: denote a 2-D representation
T: denote a 3-D representation
Subscript 1 or 2 denote $\chi = +1$ or -1 under the $C_2(\perp C_n)$ or σ_V (if no $C_2(\perp C_n)$).
" ′" and """ denote $\chi = +1$ or -1 under the σ_h.
g and u denote $\chi = +1$ or -1 for i

A full character table of C_{3v} point group

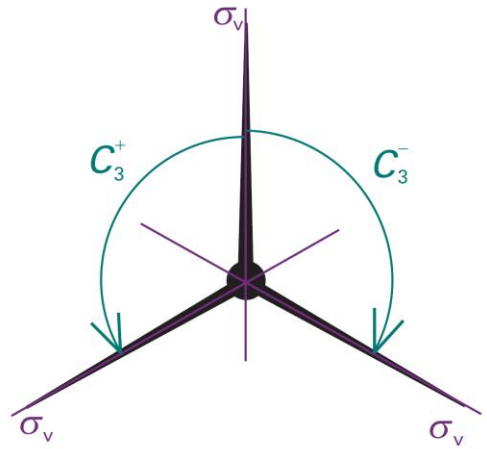
Order of a group = # of operations

C_{3v}	E	$2C_3$	$3\sigma_v$	$h=6$	
A_1	1	1	1	z	z^2, x^2+y^2
A_2	1	1	−1	Basis	
E Degeneracy is 2	2	−1	0	(x, y)	$(xy, x^2-y^2), (yz, zx)$

of members of a class

➡ # of irreducible representations = # of class of the symmetry operations

Symmetry operations in the same class are related to one another by the symmetry operations of the group



➡ $\sum(\text{dimensions of } irrep.)^2 = h$ $\quad C_{3v}: 1^2 + 1^2 + 2^2 = 6$

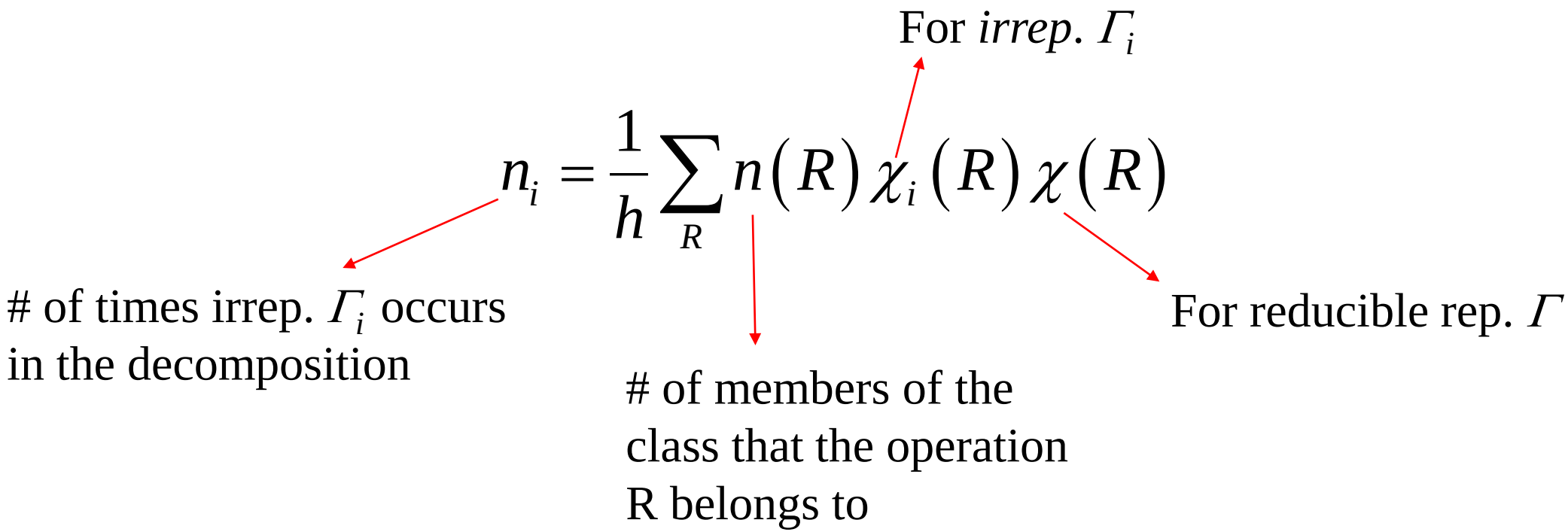
➡ $$\frac{1}{h}\sum_R \chi_i(R)\chi_j(R) = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

The character of the operation R in the ith and j^{th} *irrep*.

Apply to practical problem:

1. Choose a basis to get a reducible representation Γ;
2. Reduce Γ to *irrep*. Γ_i;
3. Understand the problem based on *irrep*.

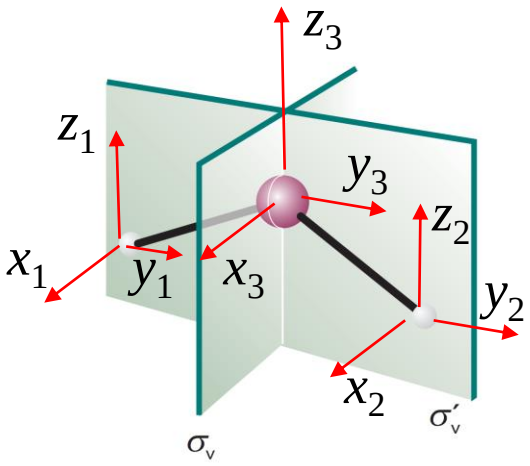
Reduction of reducible representation



Vibrational spectrum of H_2O

$$n_i = \frac{1}{h}\sum_R n(R)\chi_i(R)\chi(R)$$

C_{2v}	E	C_2	σ_v	σ'_v	$h=4$		
A_1	1	1	1	1	z		z^2, y^2, x^2
A_2	1	1	−1	−1		R_z	xy
B_1	1	−1	1	−1	x	R_y	zx
B_2	1	−1	−1	1	y	R_x	yz
Γ	**9**	**−1**	**1**	**3**			



Take 9 degrees of freedom as a basis

$$\Gamma = 3A_1 + A_2 + 2B_1 + 3B_2$$

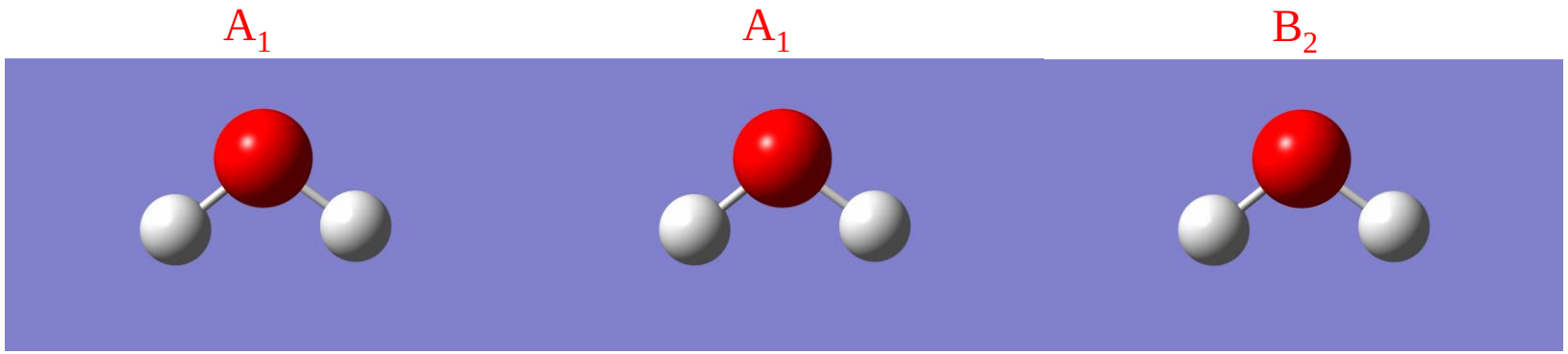
Translations	A_1	+ B_1	+ B_2
Rotations	A_2	+ B_1	+ B_2
Vibrations	$2A_1$		+ B_2

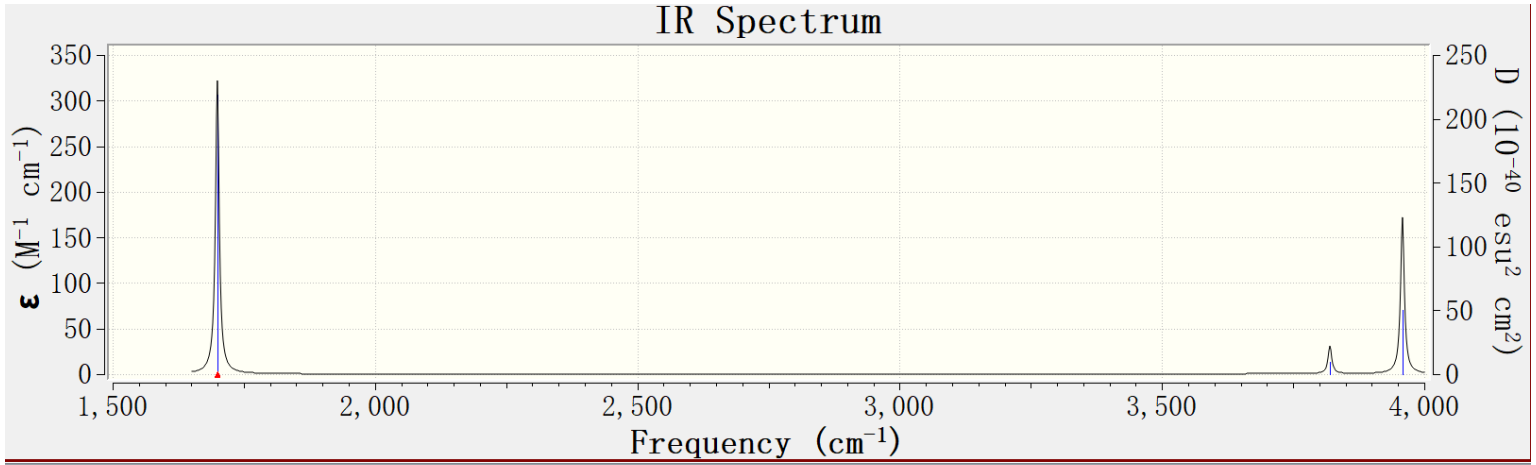
IR active:
with symmetry same as *x, y, or z*

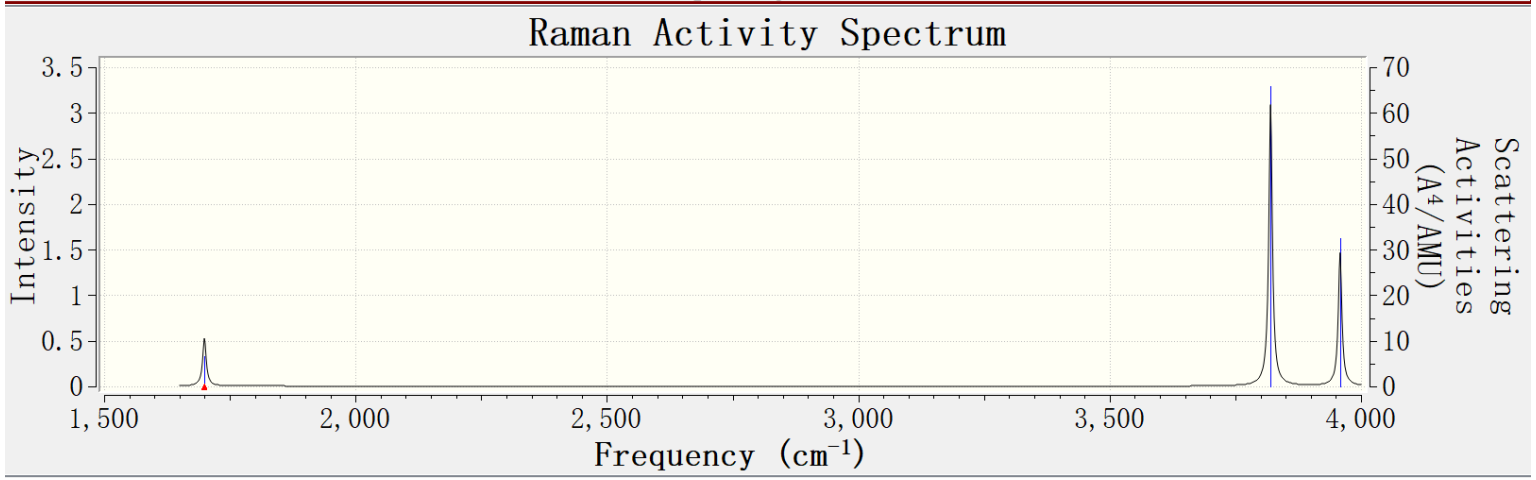
Raman active:
with symmetry same as x^2, y^2, z^2, *xy, xz, yz, or* x^2-y^2

The three vibrations are both IR and Raman active!

Vibrational spectrum of H_2O





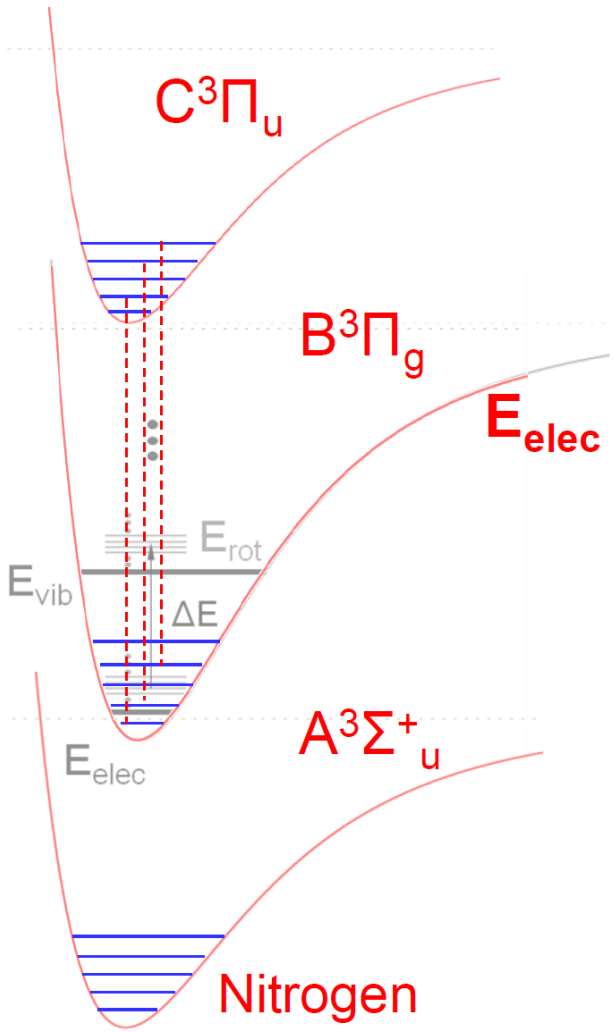


Chapter 2

Molecular spectroscopy

Electronic transitions

Electronic transitions



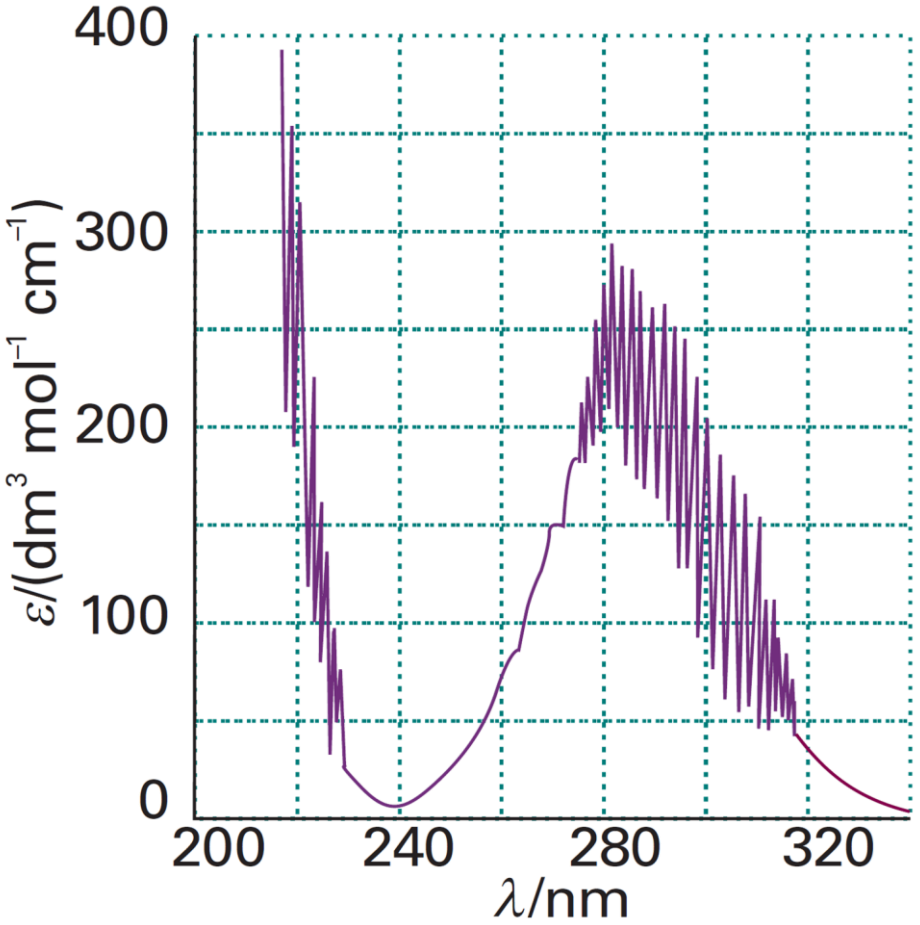
Change of electronic distributions;
Visible and ultraviolet regions

$$E_{int} = E_{elec} + E_{vib} + E_{rot} \quad (\Delta E_{rot} < \Delta E_{vib} < \Delta E_{elec})$$

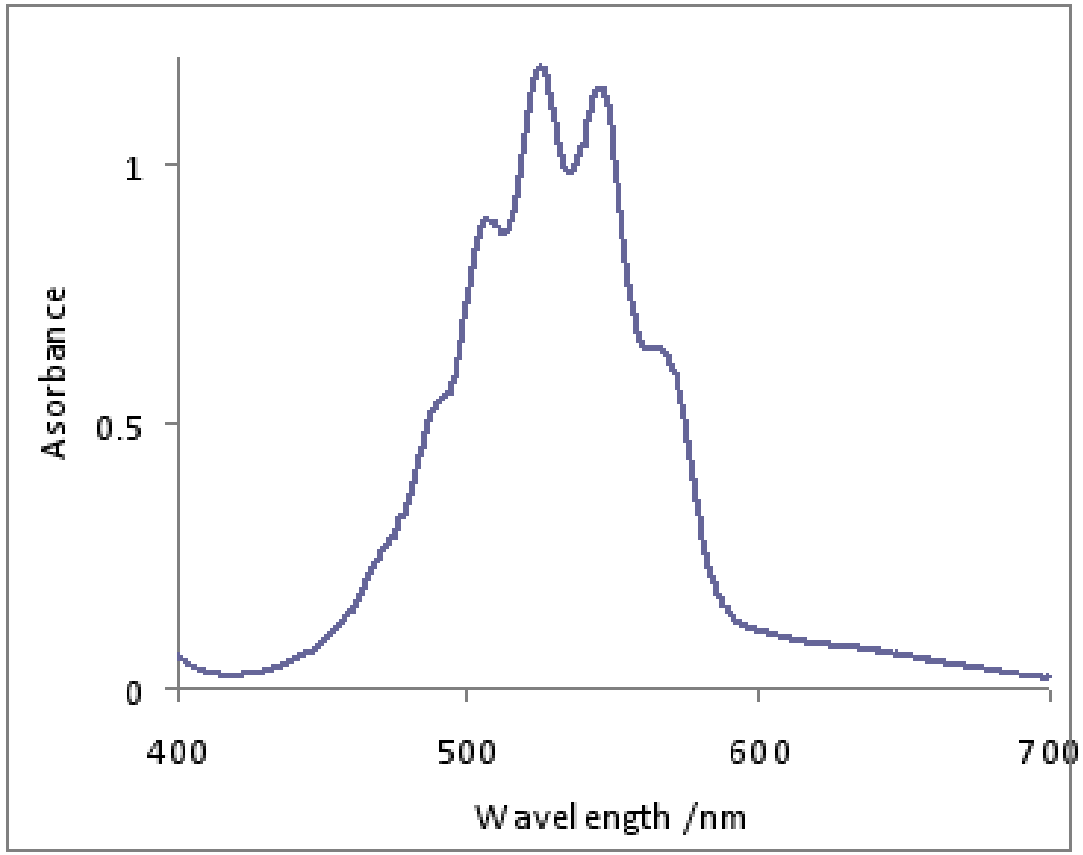
Rotational	Rovibrational	Rovibronic
Microwave Region (ΔJ)	Infrared Region (Δv, J)	UV & Visible Region (ΔΛ, v, J)
Rotation	Rotation + Vibration	Rotation + Vibration + Electronic
	Rot-Vib interaction	→ Spin-Orbital coupling
Diatomic → Polyatomic	Diatomic → Polyatomic	Single electron → Multi-electron atom → Molecules

Electronic transitions

Gaseous SO_2



Usually broad, featureless band in solution



The electronic spectra with vibrational structure

Electronic Terms of Diatomic Molecules

Atomic terms

$^{2S+1}L_J$

Atomic subshell:
each set of degenerate AOs

$2p_{+1}, 2p_0, 2p_{-1} \Rightarrow 2p$ subshell

$L = l_1 + l_2, l_1 + l_2 - 1, \ldots, |l_1 - l_2|$

$S = s_1 + s_2, s_1 + s_2 - 1, \ldots, |s_1 - s_2|$

$J = L + S, L + S - 1, \ldots, |L - S|$ *for light atoms*

L:	0	1	2	3	4
	S	P	D	F	G

Molecular terms

$^{2S+1}\Lambda_\Omega \quad \Lambda \equiv \left|\sum_i m_{l,i}\right| \quad S \equiv \left|\sum_i m_{s,i}\right|$

i, individual electron
$m_{s,i} = \pm 1/2$

Molecular subshell:
each set of degenerate MOs $\quad \pi_u(2p_{+1}), \pi_u(2p_{-1}) \Rightarrow \pi_u(2p)$ subshell

σ shell: $m_l = 0$; π shell: $m_l = \pm 1$; δ shell: $m_l = \pm 2$

Λ	0	1	2	3	4
letter	Σ	Π	Δ	Φ	Γ

$M_L = +\Lambda, \ -\Lambda$

$M_S = S, S-1, \ldots, -S$

$\Omega = \Lambda + S, \Lambda + S\text{-}1, \ldots, |\Lambda - S|$

The component of electronic orbital angular momentum along the molecular axis has the possible values $M_L\hbar$

The component of ***S*** along an axis has the possible values $M_S\hbar$

A closed-shell molecular configuration has $S = \Lambda = 0 \Rightarrow$ only a $^1\Sigma$ term

The terms of molecules with open shell(s) are determined by the electrons in the open shell(s)

Molecule H_2^+

Configuration $(1\sigma_g)^1$

↑ or ↓

$m_l = 0$

$m_s = \pm 1/2$

$\Lambda = 0$, $S = 1/2$

$^{2S+1}\Lambda$

$$\Lambda \equiv \left| \sum_i m_{l,i} \right|$$

$$S \equiv \left| \sum_i m_{s,i} \right|$$

$2S+1$

$^2\Sigma_g^+$

Λ :	0	1	2
	Σ	Π	Δ

g × **g** = **g**
g × **u** = **u**
u × **u** = **g**

Parity
奇偶性

eigenvalue +1 with respective to reflection of electronic coordinates in xz symmetry plane containing the molecular axis;
"–" refers to eigenvalue – 1

Electronic Terms

$^{2S+1}\Lambda$

$$\Lambda \equiv \left|\sum_i m_{l,i}\right|$$

$$S \equiv \left|\sum_i m_{s,i}\right|$$

σ shell: $m_l = 0$;
π shell: $m_l = \pm 1$;
δ shell: $m_l = \pm 2$

Λ	0	1	2	3	4
letter	Σ	Π	Δ	Φ	Γ

g × g = g
g × u = u
u × u = g

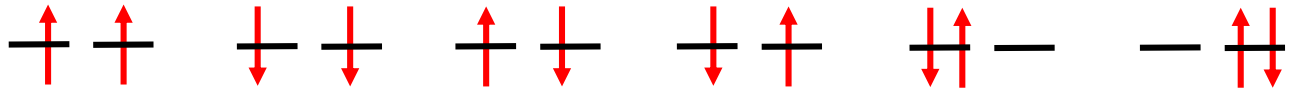
A given set of **Λ** and ***S*** include $2(2S+1)$ (if $\Lambda \neq 0$) or $(2S+1)$ (if $\Lambda = 0$) degenerate eigenstates!

Nonequivalent electron

Equivalent electron

Configuration	$m_{l,i}$	$m_{s,i}$	Λ	*S*	Terms
$\sigma^1\,\sigma^1$	0,0	±1/2	0	0, 1	$^1\Sigma$, $^3\Sigma$
$\sigma^1\,\pi^1$	0, ±1	±1/2	1	0, 1	$^1\Pi$, $^3\Pi$
$\pi^1\delta^1$	±1, ±2	±1/2	1,3	0, 1	$^1\Pi$, $^3\Pi$, $^1\Phi$, $^3\Phi$
$\pi^1\pi^1$	±1, ±1	±1/2	0, 2	0, 1	$^1\Sigma^+$, $^3\Sigma^+$, $^1\Sigma^-$, $^3\Sigma^-$, $^1\Delta$, $^3\Delta$
π^2	±1, ±1	±1/2	0, 2	0, 1	$^1\Sigma^+$, $^3\Sigma^-$, $^1\Delta$,

O_2, π_g^2 has in total $C_4^2 = 6$ microstates.



m_l	+1 -1	+1 -1	+1 -1	+1 -1	+1 -1	+1 -1
M_L =	0	0	0	0	2	-2
M_S =	1	-1	0	0	0	0
	$^3\Sigma_g^-$	$^3\Sigma_g^-$	$^3\Sigma_g^-$	$^1\Sigma_g^+$	$^1\Delta_g$	$^1\Delta_g$

Selection Rules

In light atoms

$\Delta S=0$; $\Delta L=0,\pm1$; $\Delta l=\pm1$;

$\Delta J=0,\pm1$, but $J=0$ to $J=0$ is forbidden

In heavy atoms

$\Delta S=\pm1$ are allowed.

For molecules

$\Delta S = 0$; $\Delta\Lambda = 0,\pm1$; $\Delta M_S = 0$

$\Delta\Omega = 0, \pm1$

Origins of these rules:

- Conservation of angular momentum;
- A photon has a spin of 1.

Selection rules concerned with symmetry

- For Σ terms, only $\Sigma^+ \leftrightarrow \Sigma^+$ and $\Sigma^- \leftrightarrow \Sigma^-$ transitions are allowed;
- For molecules with a center of inversion and atoms,
  u $\rightarrow$ g and g $\rightarrow$ u are allowed, but g $\rightarrow$ g and u $\rightarrow$ u are forbidden.

Some immediate consequences of symmetry

- *Only molecules with C_n , C_{nv}, and C_s symmetries may have a permanent electric dipole moment.*
- *A molecule may be chiral, if it does not have a center of inversion or a mirror pane, or an S_4 axis.*

A full character table of C_{3v} point group

Order of a group = # of operations

C_{3v}	E	$2C_3$	$3\sigma_v$	$h=6$	
A_1	1	1	1	z	z^2, x^2+y^2
A_2	1	1	−1	Basis	
E Degeneracy is 2	2	−1	0	(x, y)	$(xy, x^2-y^2), (yz, zx)$

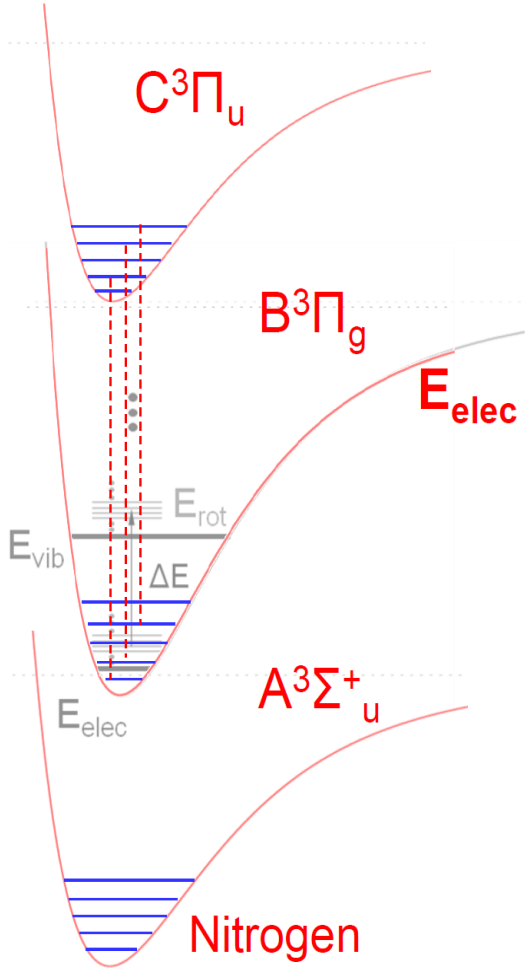
of members of a class

Apply the character table to practical problem:

1. Choose a basis to get a reducible representation Γ;
2. Reduce Γ to *irrep*. Γ_i;
3. Understand the problem based on *irrep*.

$$n_i = \frac{1}{h}\sum_R n(R)\chi_i(R)\chi(R)$$

$E_{int} = E_{elec} + E_{vib} + E_{rot}$ $(\Delta E_{rot} < \Delta E_{vib} < \Delta E_{elec})$



Electronic terms of diatomic molecules

$$^{2S+1}\Lambda_{\Omega} \qquad \Lambda \equiv \left|\sum_i m_{l,i}\right| \qquad S \equiv \left|\sum_i m_{s,i}\right|$$

σ shell: $m_l = 0$; π shell: $m_l = \pm 1$; δ shell: $m_l = \pm 2$

$$M_L = +\Lambda, \ -\Lambda$$
$$M_S = S, S-1, \ldots, -S$$

Λ	0	1	2	3	4
letter	Σ	Π	Δ	Φ	Γ

$$\Omega = \Lambda + S, \ \Lambda + S\text{-}1, \ldots, \ |\Lambda - S|$$

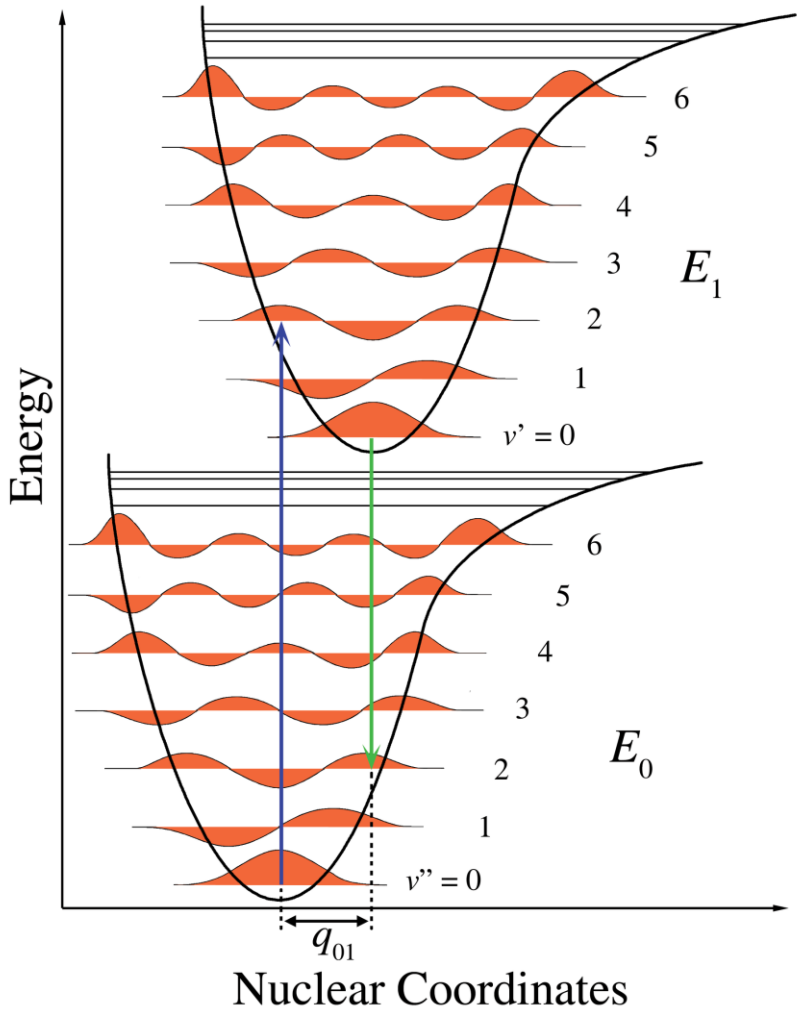
Selection rules of electronic transitions for molecules

$\Delta S = 0$; $\Delta\Lambda = 0, \pm 1$; $\Delta M_S = 0$; $\Delta\Omega = 0, \pm 1$

Selection rules concerned with symmetry

- For Σ terms, only $\Sigma^+ \leftrightarrow \Sigma^+$ and $\Sigma^- \leftrightarrow \Sigma^-$ transitions are allowed;
- For molecules with a center of inversion and atoms, u → g and g → u are allowed, but g → g and u → u are forbidden.

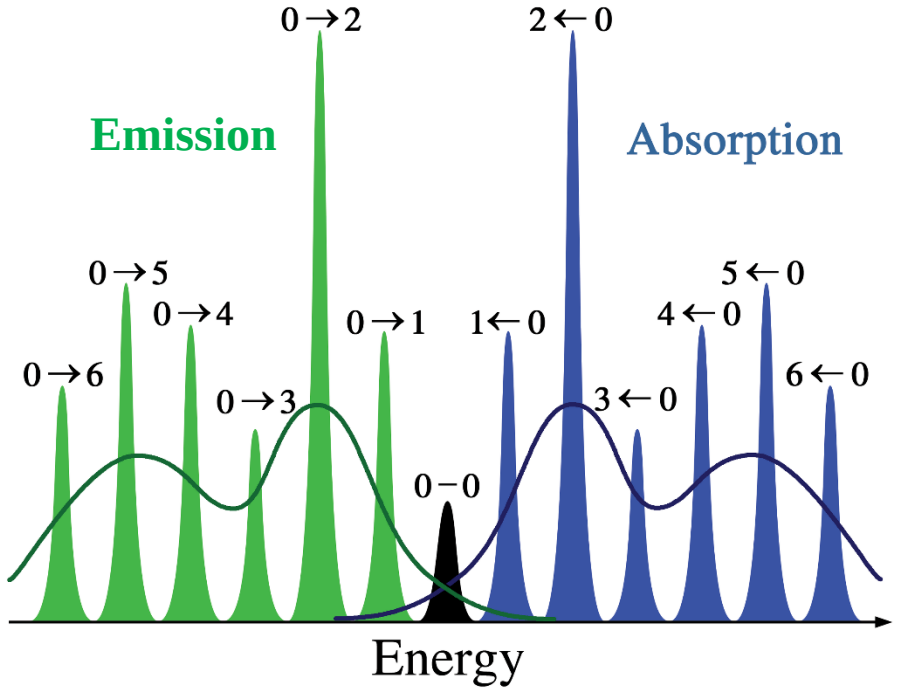
Electronic transitions



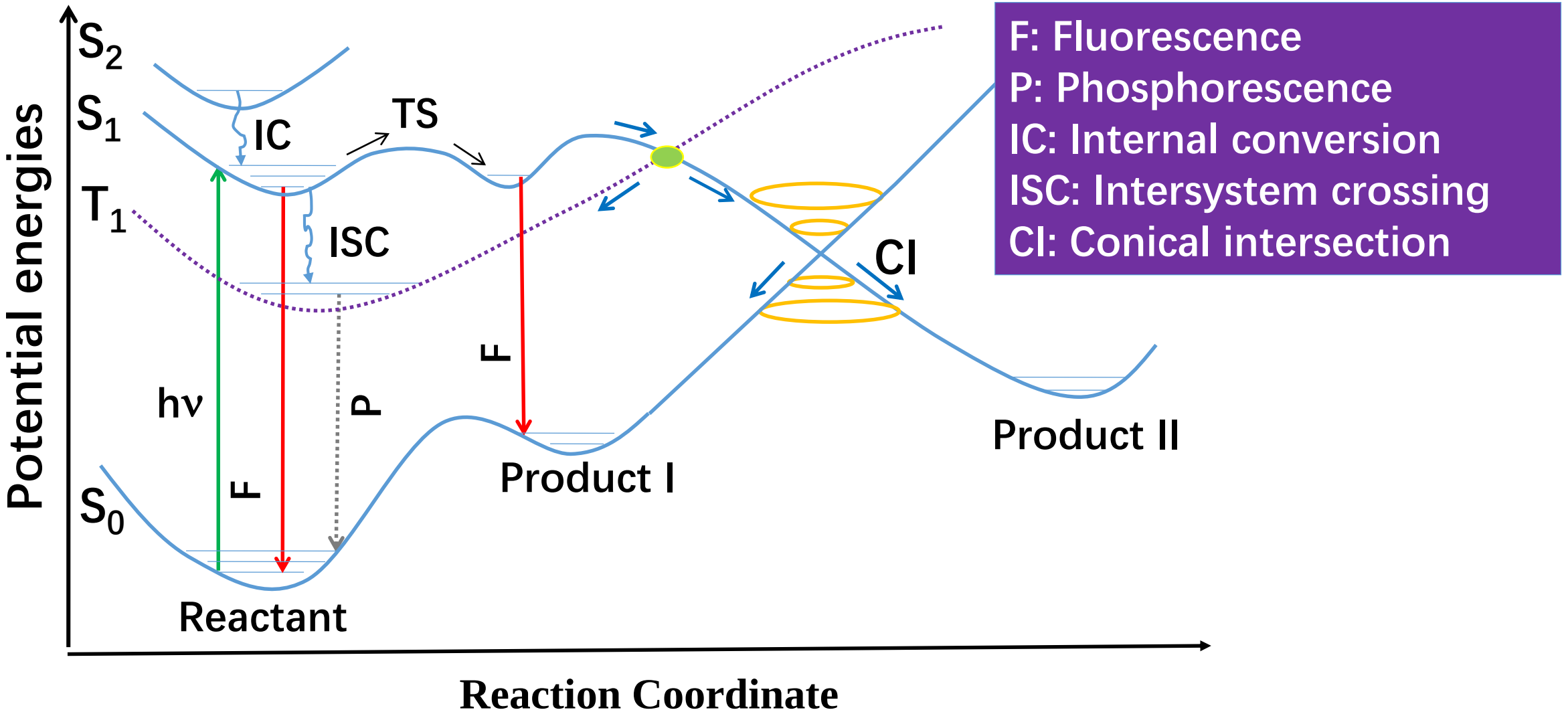
Franck–Condon principle: Because the nuclei are so much more massive than the electrons, an electronic transition takes place very much faster than the nuclei can respond.

Vertical transition: An electronic transition that occurs without change of nuclear geometry.

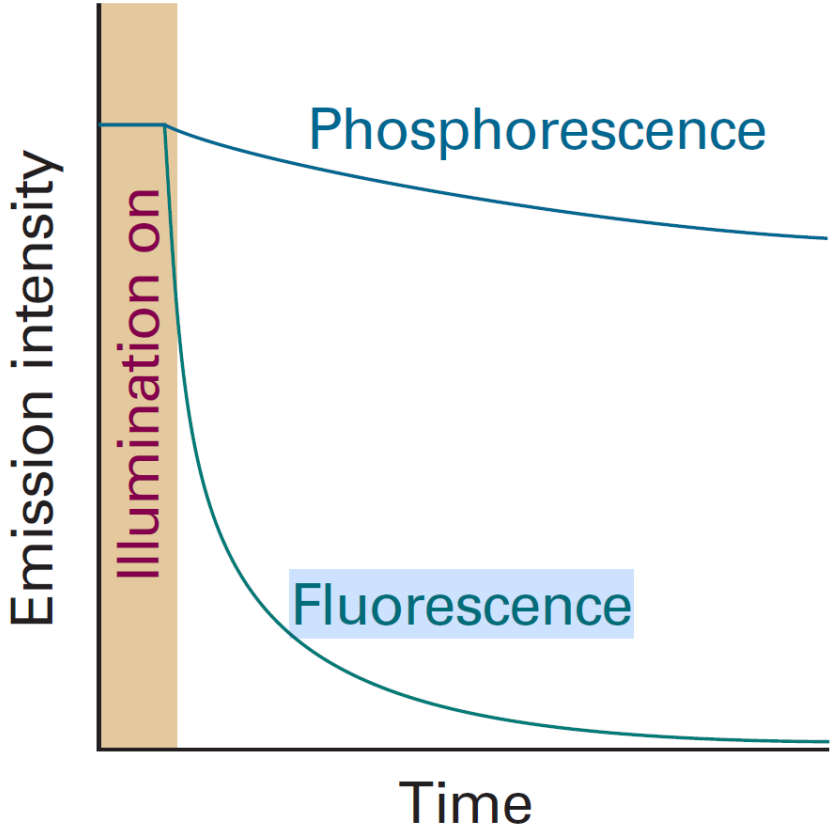
0-0 transition: Electronic transitions between the lowest vibrational levels of the electronic states.



The fates of electronically excited states



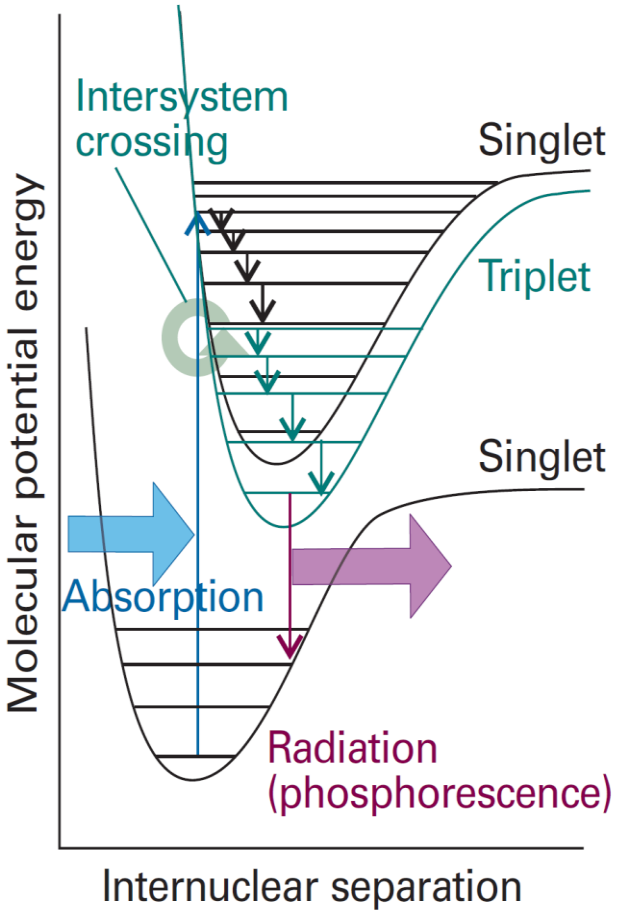
The fates of electronically excited states

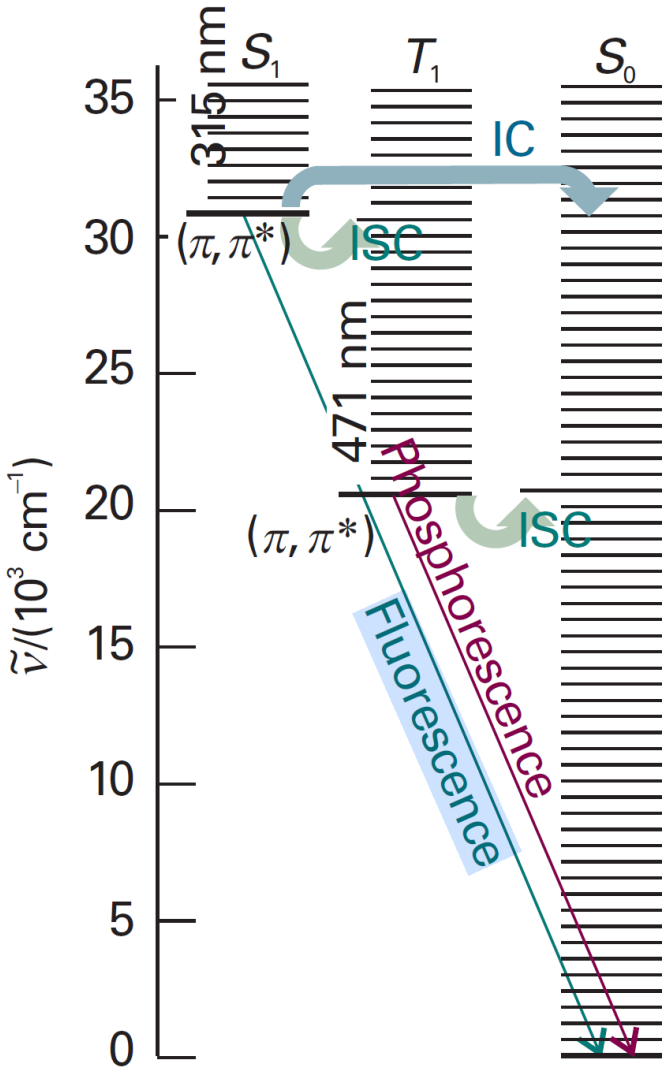


Fluorescence (10^{-12}–10^{-6} s):
Spontaneous emission of radiation occurs within a few nanoseconds after the exciting radiation is extinguished.

Phosphorescence (10^{-6}–10^{-1} s):
Spontaneous emission may persist for long periods.

The fates of electronically excited states





A Jablonski diagram for naphthalene

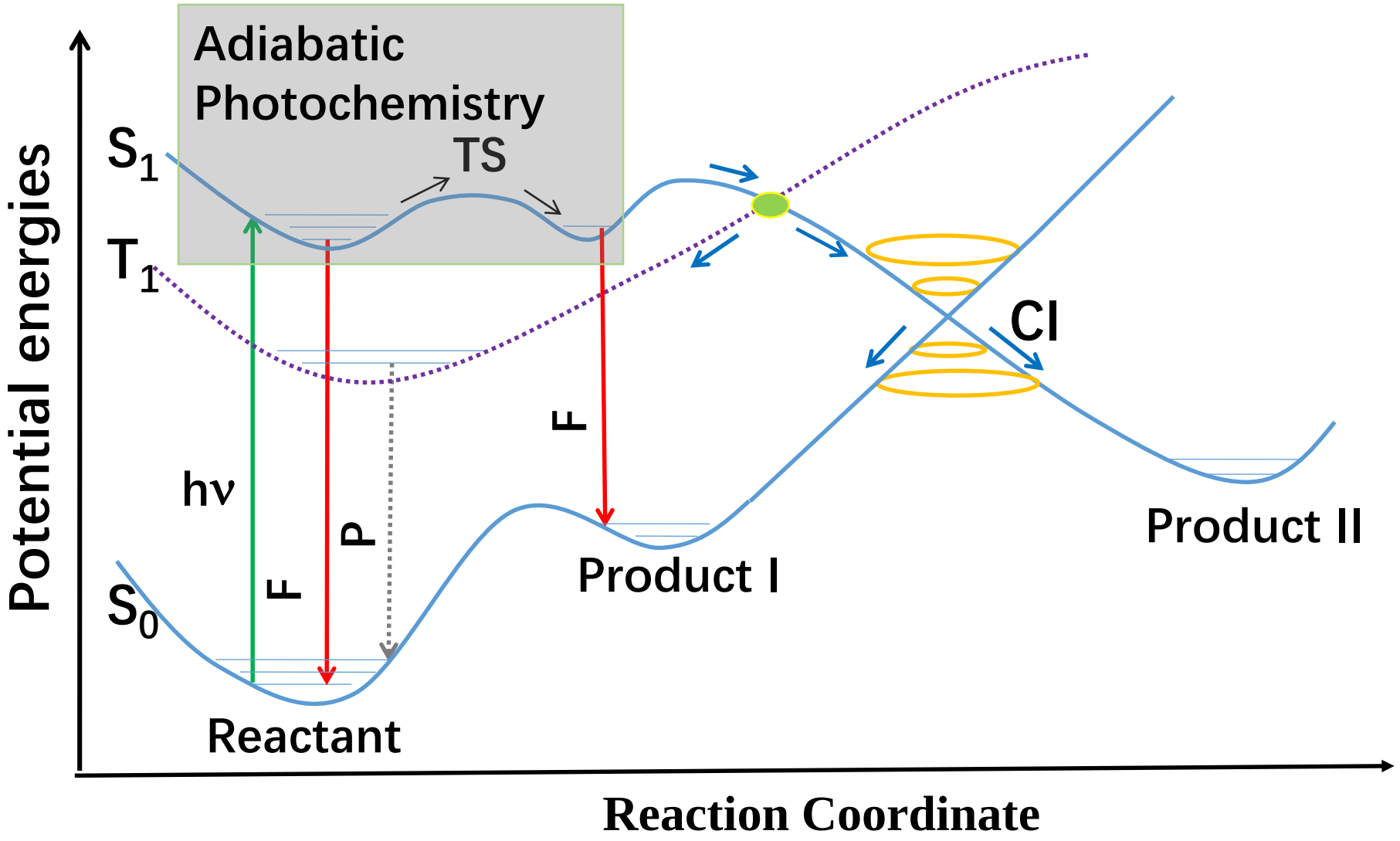
Intersystem crossing:

- A nonradiative transition between states of different multiplicity;
- May occur in the presence of spin-orbit coupling;
- Important when contain a moderately heavy atom.

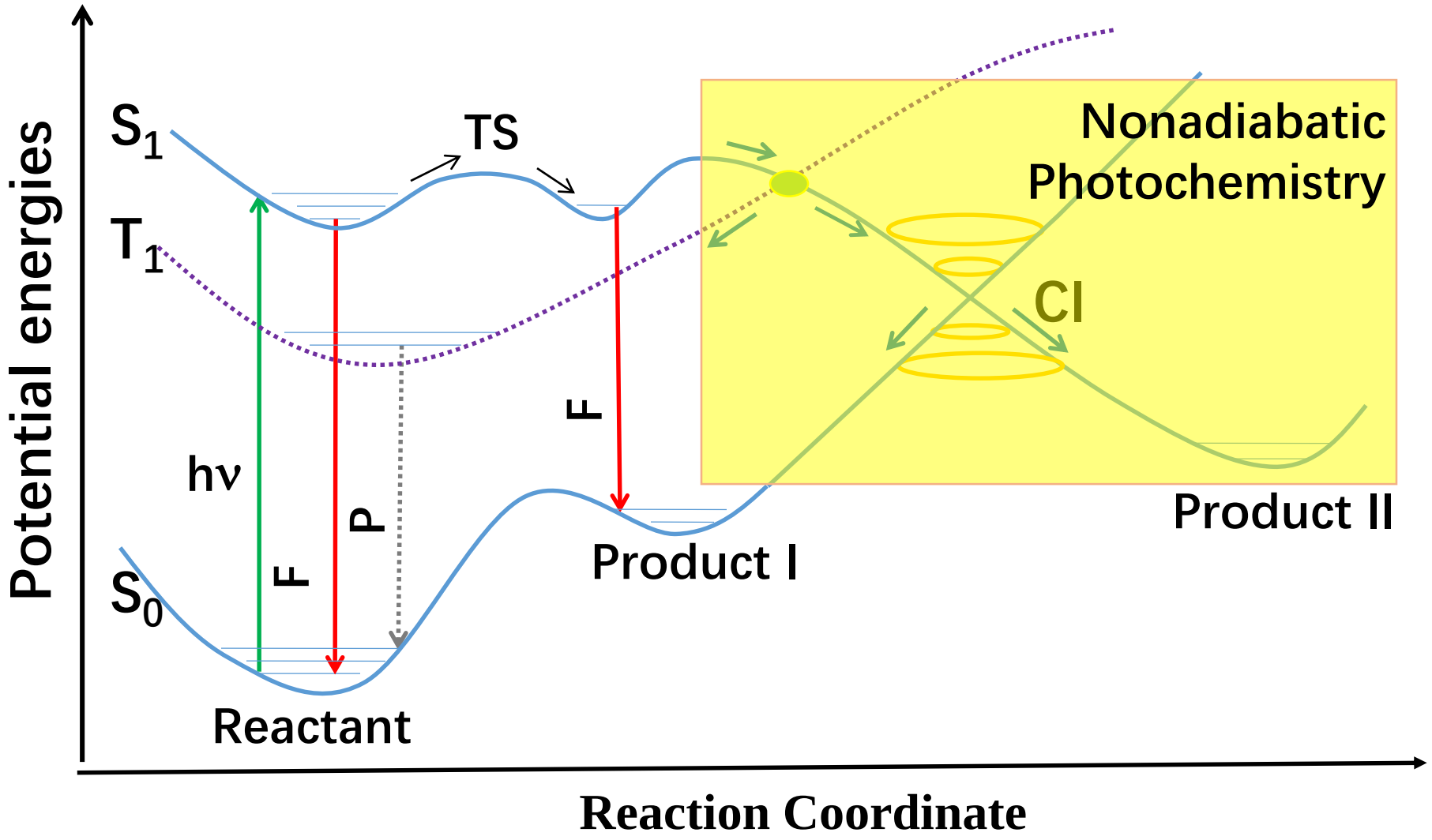
Internal conversion:

- A nonradiative transition between states with the same multiplicity.

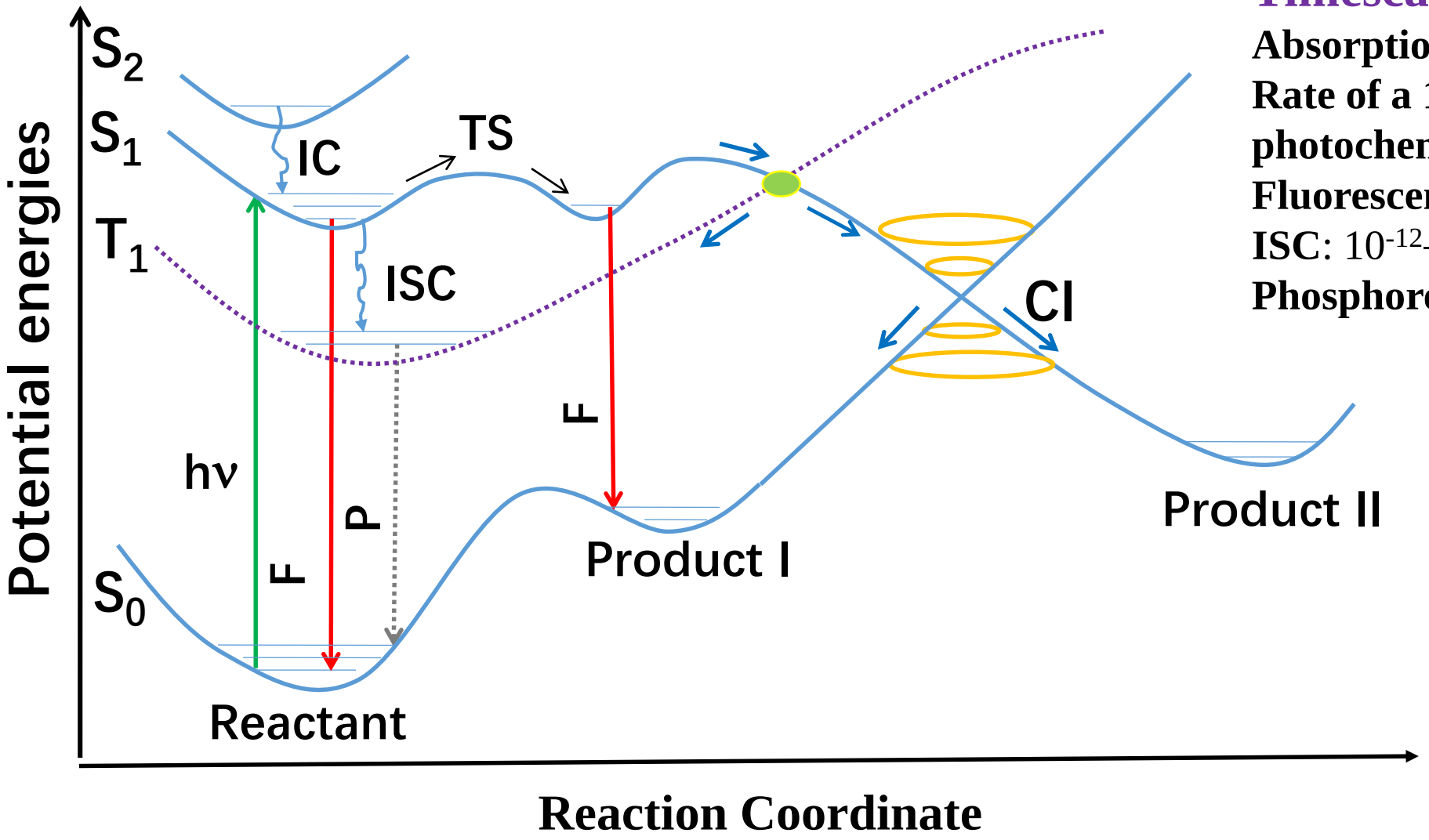
The fates of electronically excited states



The fates of electronically excited states



The fates of electronically excited states



Timescales:

Absorption: 10^{-16}–10^{-15} s

Rate of a 1st-order photochemical reaction: 10^{16} s^{-1}

Fluorescence: 10^{-12}–10^{-6} s

ISC: 10^{-12}–10^{-4} s

Phosphorecence: 10^{-6}–10^{-1} s